

Project 21: Angus & Coote Online Jewellery Store Management System

About the Client

Angus & Coote is Australia's industry leader in jewellery, watch and giftware specialty retailing.

Formed in Sydney in 1895, the Company has been a leading name in Australian jewellery ever since and has continued to enjoy a reputation for consistently high quality, value and service.

Angus & Coote offers customers a comprehensive range of stunning diamonds, coloured gems, gold and silver jewellery, watches, giftware and related products.

Ever since its earliest days, Angus & Coote has been a home for superb craftsmanship. Today, the tradition of craftsmanship continues, thanks to a wide range of suppliers from around the world.

The long-term success of Angus & Coote rests on our key values, including dedication to the customer, assurance of quality and value, uniqueness and innovation, excellence, leadership and responsibility. For you, this means ensuring you receive a piece you can be proud of for life.

Furthermore, you can rest assured that Angus & Coote will still be here for many years to come. If you need an alteration or repair, the Company is well placed to assist. Throughout its history, the Company has remained true to its founding principles, while evolving to reflect changing demands. When you purchase an item of jewellery from Angus & Coote, you can be confident you're taking part in an Australian tradition.

Project Brief and Business Problem Specifications:

Our client Angus & Coote wants to completely redesign their online presence with new tech implementation on both frontend and backend levels. The existing system need to redesign due to the following reasons:

Higher Overall Traffic. With more visitors coming to the website, client will have more chances to acquire new customers.

Higher Conversion Rates. An attractive website and storefront will draw in more visitors and encourage them to make a purchase.

More Leads and Sales. An e-commerce website redesign should optimize the shop to match the digital marketing strategies, pulling in more leads and closing more sales.

Better Customer Retention. The experience of shopping on a top-tier website will inspire many of the customers to return to the shop and make additional purchases.

The project should be a web-based application with 150 products store and extendable in future. The main goal of this project is to create a shopping cart, which allows customers to shop and purchase Jewelry products online.

Moreover, the system should be a web-based e-commerce application that allows customers to browse, try and purchase jewellery items online using Augmented Reality (AR) technology. The system should provide an immersive experience by enabling users to virtually 'try on' rings, necklaces, earrings and bracelets using their webcam or smartphone camera before purchasing.

The platform should integrate a PHP based backend and MySQL database for product, user and order management while WebAR / AR.js / Three.js should handle the AR interaction layer.

System Modules Requirements

The application should consist of following main components:

- User Management Module
- Product Catalogue Module
- AR Try-On Module
- Shopping Cart and Checkout Module
- Payment and Order Management Module
- Admin Management Module
- AI Recommendation and Chatbot Module

User Modules (User Frontend):

Developers need to research and discuss with the client to finalise the modules and requirements.

UI Design

User Interface Design is concerned with the dialogue between a user and the computer. It is concerned with everything from starting the system or logging into the system to the eventual presentation of desired inputs and outputs. The overall flow of screens and messages is called a dialogue.

UI Design Requirements

1. The system user should always be aware of what to do next.
2. The screen should be formatted so that various types of information, instructions and messages always appear in the same general display area.
3. Message, instructions or information should be displayed long enough to allow the system user to read them.
4. Use display attributes sparingly.
5. Default values for fields and answers to be entered by the user should be specified.
6. A user should not be allowed to proceed without correcting an error.
7. The system user should never get an operating system message or fatal error.

The aim of proposed system is to develop a system of improved facilities. The proposed system can overcome all the limitations of the existing system. The system provides proper security and reduces the manual work.

- Security of data.

- Ensure data accuracy's
- Proper control of the higher officials.
- Minimize manual data entry.
- Minimum time needed for the various processing.
- Greater efficiency. -
Better service.
- User friendliness and interactive.
- Minimum time required.

Functional Requirements:

1. User Module

1. Users should be able to register and log in using email and password.
2. Users should be able to reset forgotten passwords via email verification.
3. Users should be able to edit their personal profile information (name, address, contact).
4. Users should be able to view their order history and track current orders.
5. Users should be able to save favourite jewellery items to a wishlist.
6. Users should be able to submit product reviews and ratings.
7. Users should receive email notifications for successful registration, purchase confirmation and shipping updates.

2. Product and Catalogue Module

8. The system should allow the admin to add, edit or delete jewellery products.
9. Each product should include attributes such as name, description, price, category, images, metal type and available sizes.
10. Users should be able to view detailed product information.
11. The system should provide dynamic image rotation or zoom for product images.
12. Products should be grouped under categories such as Rings, Earrings, Necklaces and Bracelets.
13. Users should be able to search products by keyword, metal, gemstone type or price range.
14. The system should recommend similar products based on the viewed item.

3. AR Virtual Try-On Module

- 15. Users should be able to activate the AR Try-On feature via their webcam or smartphone camera.
- 16. The system should overlay 3D jewellery models (e.g. rings, necklaces, earrings) on live camera feed.
- 17. Users should be able to rotate, resize and reposition the jewellery model virtually.
- 18. The system should detect and map facial or hand landmarks for realistic alignment.
- 19. Users should be able to capture and save snapshots of their virtual try-on.
- 20. The system should support both marker-based (QR/AR marker) and markerless AR experiences.
- 21. The AR system should adjust lighting and shadows to enhance realism.

4. Shopping Cart and Checkout Module

- 22. Users should be able to add jewellery items to their shopping cart.
- 23. Users should be able to update quantity or remove items from the cart.
- 24. The system should calculate total cost, tax and shipping charges automatically.
- 25. The checkout process should allow users to confirm delivery address and payment method.
- 26. Users should be able to apply discount coupons or promotional codes.
- 27. The system should display an order summary before confirmation.
- 28. Users should receive an invoice in PDF format after successful payment.

5. Payment and Order Management Module

- 29. The system should integrate with online payment gateways (e.g. PayPal, Stripe).
- 30. The system should record transaction details and update order status automatically.
- 31. Users should be able to view order status as Pending, Confirmed, Shipped or Delivered.
- 32. The system should send order confirmation and shipping emails automatically.
- 33. Admin should be able to update order statuses manually from the dashboard.

6. Admin Module

- 34. Admin should be able to manage product inventory (add, edit, delete).

- 35.Admin should be able to view, search, and manage all registered users.
- 36.Admin should be able to manage orders, view sales reports and update shipping details.
- 37.Admin should be able to add promotional banners, discount codes and featured products.
- 38.Admin should be able to view feedback and customer reviews.
- 39.Admin should be able to upload new 3D AR jewellery models.
- 40.Admin should be able to generate monthly sales and customer analytics reports.

7. AI and Recommendation Module (Optional Enhancement)

- 41.The system should analyse user browsing and purchase history to suggest personalised jewellery items.
- 42.The system should provide AI-based virtual fitting suggestions (e.g. best ring size or necklace length).
- 43.The system should generate automatic descriptions for new jewellery using natural language templates.
- 44.The AI chatbot should assist customers with product selection and FAQs.
- 45.The system should learn from user interactions to improve recommendation accuracy.

Non-Functional Requirements:

1. Performance

- The system should load the main product catalogue within 3 seconds under normal network conditions.
- The AR rendering should maintain ≥ 25 FPS on standard mobile browsers.
- The database queries should execute within 1 second for most transactions.

2. Security

- Passwords should be encrypted using SHA-256 or bcrypt.
- User sessions should automatically expire after 15 minutes of inactivity.
- All transactions should occur over HTTPS.
- Input validation should prevent SQL injection, XSS and CSRF attacks.
- Payment gateway integration should follow PCI-DSS standards.

3. Usability

- The system should have a responsive UI compatible with desktop and mobile devices.
- Navigation menus and icons should follow e-commerce usability best practices.
- AR try-on instructions should be clearly displayed to guide first-time users.

4. Reliability and Availability

- The system should maintain 99% uptime during operational hours.
- Database backup should occur automatically every 24 hours.
- The system should recover from server crashes within 5 minutes using backup replication.

5. Maintainability

- The system should use MVC architecture for PHP (e.g. Laravel or Core PHP structure).
- Code modules should be well-commented for future scalability and debugging.
- Admin should be able to update content and products without developer intervention.

6. Portability

- The AR module should be compatible with Chrome, Safari, and Firefox browsers.
- The application should run on Apache web servers.

7. Scalability

- The system should support up to 10,000 registered users without major performance degradation.
- The architecture should allow future integration of VR (Virtual Reality) modules.

8. Data Integrity

- The system should maintain referential integrity between user, product and order tables.
- Any transaction failure should roll back automatically to prevent partial updates.

9. Accessibility

- The system should comply with WCAG 2.1 guidelines for accessibility.
- Alternative text should be provided for all images and AR markers.

Recommended Tech Stack by the Client:

Frontend: HTML5, CSS3, JavaScript, Bootstrap, AR.js / Three.js

Backend: PHP (Core or Laravel)

Database: MySQL

AR Framework: WebXR / AR.js / 8thWall (if premium integration used)

Web Server: Apache

AI Integration: TensorFlow.js or pre-trained API (e.g. Google Vision, Chatbot API)

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.